

ForeDance: Structured Streaming Dance Generation with Anticipatory Music Conditioning

Abstract

Streaming dance must prepare movements for music that has not yet arrived while continuing from an irrevocable motion prefix. This couples musical anticipation with the problem of generating coherent motion over successive decisions. We present ForeDance, a streaming dance framework that obtains forward-looking conditions from a frozen causal music model. Forecast states guide a discrete structural planner and a continuous-detail generator, providing musical lookahead without access to future audio. To make these motion components complementary, we introduce kinematics-guided structure–detail decoupling together with its training objectives: structural reconstruction preserves designated body organization and endpoint geometry, while full reconstruction retains the complete motion. Commit Forcing trains continuation from jointly committed structure and detail and the boundary state reconstructed from their decoded movement. On FineDance, forecast-conditioned models reduce a frozen music–motion matching distance relative to matched no-music training by 0.782 and 1.123 under two startup settings, with positive paired 95% intervals and consistent directions across three training seeds. These results establish that causal music forecasts provide usable conditioning within the structured streaming framework.

1 Introduction

A dance movement often begins before its most visible accent: weight shifts prepare a step, and the body transitions toward a pose before reaching it. For a system generating dance as music arrives, the next movement must therefore be chosen with incomplete knowledge of the music it will accompany. Waiting for that music sacrifices the opportunity to prepare; using the actual future recording changes the task into offline generation. The central problem is how to obtain useful musical lookahead while respecting a causal audio stream.

Music-to-dance models have made substantial progress in choreography and expressive motion [16, 17, 27, 29]. Streaming introduces an additional constraint: the model cannot revise movement already committed to its output. A new musical condition must be translated into a continuation of that movement, rather than a fresh, independent dance segment. In robot-native motion, the committed prefix also determines the pose and velocity at the next segment boundary. Anticipating the music and continuing the body are thus coupled aspects of the same generation problem.

We address this problem with ForeDance, a frame-

work for *anticipatory music conditioning in streaming dance generation*. A frozen generative music model extrapolates from the audio heard so far. Its forecast states condition upcoming motion at their corresponding physical times. This gives the dance generator a hypothesis about the music ahead without exposing it to future waveform samples. The generator repeatedly plans a short horizon, commits a prefix, and replans as more music arrives (Fig. 1). A forecast can therefore guide preparation without fixing the entire future dance.

The usefulness of this interface depends on how motion is organized. Discrete codes provide a compact vocabulary of movement, while continuous residuals can retain variation within that vocabulary. A generic combination, however, leaves their division of labor unspecified. Our *kinematics-guided structure–detail decoupling* couples the representation to its training: discrete reconstruction is supervised on root, leg, and waist quantities together with torso, wrist, and sole geometry; full reconstruction is supervised on the complete motion. These views prioritize body organization while retaining complementary articulation. The geometry constraint also connects the roles: wrist placement belongs to the structural view even though arm-joint articulation belongs to the complementary native-coordinate view. A structural planner and a structure-conditioned detail generator reunite through a shared decoder.

A second challenge arises when the model continues from its own output. Teacher Forcing (TF) trains on recorded histories and their recorded boundary states. During generation, both instead depend on the preceding sampled motion. Replacing the history alone leaves the starting state disconnected from the movement that produced it. Commit Forcing constructs the complete generated transition: it samples structure and detail, commits their shared prefix, and reconstructs the boundary state from the decoded motion. The next plan is then trained under this history–state pair. This connects the training context to the same commitment operation used at inference.

Our contributions are:

- **Anticipatory music conditioning** supplies forecast states inferred from heard audio to both motion generators, enabling musical lookahead within a causal streaming interface.
- **Kinematics-guided structure–detail decoupling and training** assign complementary motion views to discrete structure and continuous detail, with coupled generation and shared decoding.
- **Commit Forcing** trains the next plan using jointly committed generated motion and its recon-

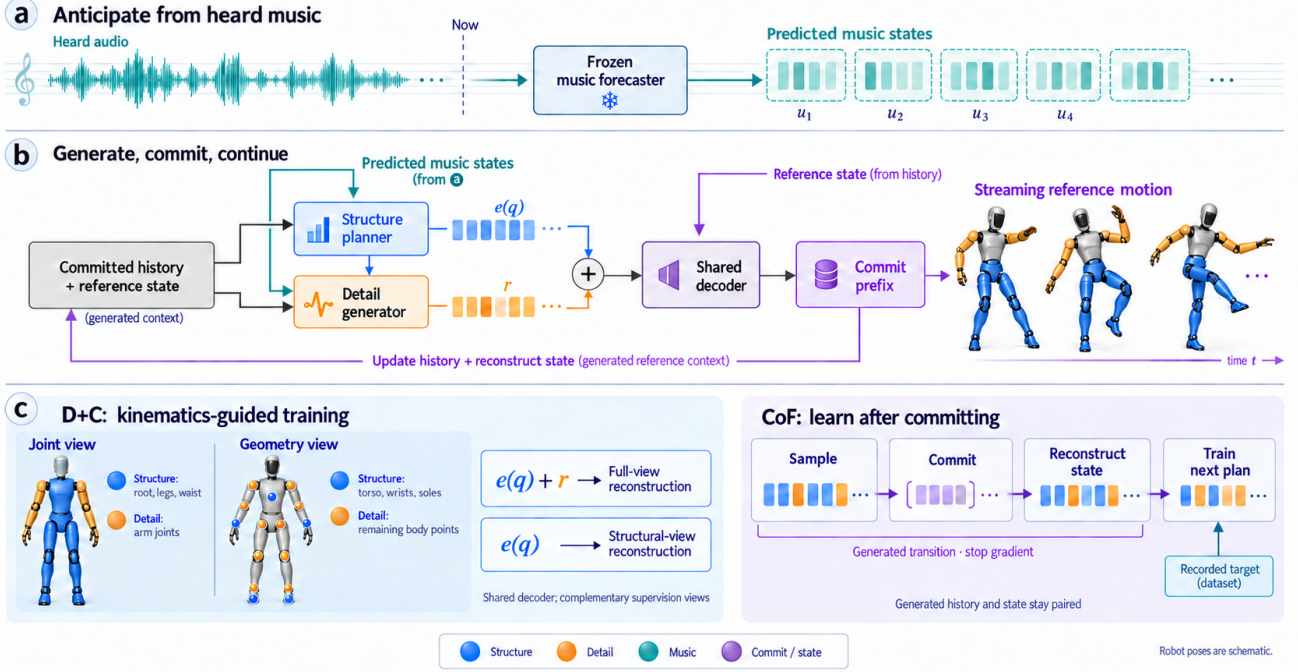


Figure 1: ForeDance overview. **Top:** heard audio supplies a causal forecast of upcoming music states. **Middle:** the forecast conditions both discrete structure and continuous detail; structure also conditions detail. Their sum passes through a shared decoder, and the committed prefix updates the motion history and reference-derived boundary state for the next decision. **Bottom:** kinematic supervision gives the two latent components complementary roles, while Commit Forcing trains continuation from a jointly generated history–state pair. Robot poses illustrate the representation and output interface; they are schematic, not experimental or hardware execution evidence.

structured boundary state, addressing the context mismatch created by successive generation decisions.

FineDance experiments establish music-condition use under both mixed-start and continuation-start training: forecast conditioning improves music–motion matching over separately trained no-music controls, with the same direction across all three training seeds. We also examine how history corruption affects the motion distribution and show a selected 60-second generation. The matched tests of forecast versus heard-music conditioning, representation alternatives, and CoF versus TF are specified separately from these completed results.

A fixed whole-body controller provides a separate execution interface for committed robot-native references. We distinguish the current generation study from supporting replay diagnostics of an earlier configuration, using the latter to explain how reference delivery and tracking are measured.

2 Related Work

Music-conditioned and streaming dance. AIST++/FACT [17] and Bailando [27] learn music-conditioned choreography, while EDGE [29] and LODGE [16] use diffusion and hierarchical generation to model expressive dance. FineDance [15] extends the data foundation to fine-grained full-body choreography. More recent work develops complementary

forms of control: MEGADance [34] uses genre-aware experts, OpenDance [37] combines music with text, keypoints, and trajectories, and SoulNet [18] couples hierarchical motion modeling with learned music–motion alignment. These capabilities concern how supplied conditions organize a dance. A streaming setting additionally asks what musical information is available before an output segment is committed. DiscoForcing [11] studies causal audio-driven character control with diffusion forcing. ForeDance focuses on inferred musical lookahead: forecast states from the heard prefix condition a structured generator that repeatedly continues from committed motion.

Generative music as a prospective condition. Jukebox [7] models compressed audio codes autoregressively, establishing a route to continuation in a learned music space. Live Music Models [24] develops continuously generated music with synchronized user control. Magenta RealTime 2 [25] and its SpectroStream codec [19] provide the causal music interface used here. Our use of these models is to expose forecast states to a downstream motion generator. The anticipated musical context is inferred from arrived audio; it does not grant access to the future recording. This distinguishes the source of the condition from the dance representation that interprets it.

Discrete structure and continuous detail. Vector quantization [30] supports discrete motion models

such as T2M-GPT [36] and MoMask [8]; latent diffusion [4] provides a continuous modeling alternative. DisCoRD [6] connects discrete tokens with continuous motion decoding, and DC-Motion [31] combines discrete structure with continuous residuals. Our contribution builds on this distinction through kinematic supervision: reconstruction views specify which joint and geometric quantities the structural code should preserve and how detail complements them. Both components retain a shared decoder. Their division of labor is therefore learned through complementary reconstruction objectives, while their coupled generation and joint commitment make the representation part of the streaming design.

Streaming generation and learning from generated context. MotionStream [13] uses a causal motion tokenizer with residual quantization for continuous text-conditioned motion, and MotionStreamer [32] combines a causal latent space with diffusion-based autoregression and Two-Forward training. More broadly, Scheduled Sampling [1] substitutes generated inputs during sequence training, Professor Forcing [14] aligns training and sampling dynamics, and Diffusion Forcing [3] trains sequences with independently varying noise levels. In video generation, Self Forcing [10] trains on autoregressive model rollouts, while Self Gradient Forcing [39] uses a second pass to supervise how generated history is encoded into causal memory. These works motivate treating generated context as a training concern. CoF addresses the specific transition in our boundary-conditioned discrete-continuous generator: it jointly commits structure and detail, decodes the prefix, and pairs the resulting code history with its reconstructed boundary state. Its first transition is sampled without gradients, and the next plan is trained under that generated history-state pair. The controlled comparison in this paper is CoF versus TF; the other training approaches establish the methodological context.

Committed prefixes and robot motion interfaces. Receding-horizon generation also appears in Diffusion Policy [5]. Real-Time Chunking [2] handles asynchronous policy execution by freezing actions scheduled to execute while inpainting the remainder of a new chunk. This relates to preserving an output prefix, whereas CoF constructs the training history and boundary state induced by committing a motion plan. Physical motion generation and tracking address complementary questions: PhysDiff [35] introduces simulator-based projection during denoising; PHC [22] tracks reference motion with recovery in simulated avatars; and ExBody2 [12] and KungfuBot [33] learn expressive or highly dynamic humanoid tracking. BeyondMimic [21] extends tracking to guided motion composition, while SONIC [23] scales generalist whole-body tracking. Most directly related to the audio interface, RoboPerform [20] conditions humanoid behavior on content and audio-derived style. ForeDance studies anticipatory conditioning and continuation in a na-

tive robot reference stream. A separate controller may track this reference; the generator’s boundary state is reconstructed from its committed output.

3 Problem Formulation

We consider a causal audio stream and an output motion prefix that cannot be revised. At a planning decision, let $a_{\leq \tau}$ denote the arrived audio, h the committed motion-code history, and b the state derived from the last committed native frames. A frozen music model produces forecast states $m^{\text{pred}} = F(a_{\leq \tau})$. The motion generator samples a plan

$$(q_{1:H}, r_{1:H}) \sim p_{\theta}(q, r \mid h, b, m^{\text{pred}}). \quad (1)$$

Here q is discrete structure and r is continuous detail. Both are decoded into native G1 motion; the first $C < H$ latent steps are committed and the suffix is discarded. The next decision uses the appended history and the state reconstructed from the newly committed frames.

The motion rate is 30 Hz, with two frames per latent step. We use 64 latent history steps (128 frames, approximately 4.27 s), an eight-step plan (16 frames, approximately 0.53 s), and a four-step commit (eight frames, approximately 0.27 s). The overlapping planning horizon permits revision of uncommitted motion while the output prefix grows. All inference inputs are available at the corresponding decision time.

4 Method

ForeDance links three operations: forecasting the upcoming musical context, generating a structured motion plan, and learning to continue after committing part of that plan. Figure 1 shows how the forecast and committed motion context meet in the two motion generators.

4.1 Anticipatory Music Conditioning

The music that will accompany the next plan is only partly observed. We obtain a forward-looking condition by tokenizing the arrived audio with SpectroStream and rolling a frozen Magenta RealTime 2 Small model forward. The condition consists of 14 native forecast states at 25 Hz, spanning approximately 0.56 s. Each state retains its intended physical time. Forecasting in this latent music space supplies a prospective condition directly to the motion generators, while the future recording remains unobserved.

Let u_j be a forecast state and ℓ_j its lead relative to the motion decision. We form

$$c_j = W \text{LN}(u_j) + \phi(\ell_j), \quad (2)$$

where ϕ represents physical time. The interface aligns valid forecast states with the motion horizon and uses frame-wise feature modulation [26] in both branches.

In schematic form, an activation v_i receives a scale and a shift derived from its aligned condition:

$$\text{FiLM}(v_i, c_i) = \gamma(c_i) \odot v_i + \beta(c_i). \quad (3)$$

Conditioning both branches lets the same musical forecast guide body organization and its complementary articulation. Timestamp-based masking excludes expired forecast states from their aligned motion positions.

The forecaster remains frozen while the dance generator learns to interpret these states. A null condition bypasses music modulation and supports separately trained no-music controls. Section 5.2 distinguishes the effect of using music from the additional question of whether forecast states improve on heard-music states.

4.2 Kinematics-Guided Structure–Detail Decoupling

The representation must retain both coordinated body motion and the articulation needed to realize it. We impose this division through kinematic reconstruction views, then use the resulting discrete and continuous codes in a coupled generator.

Native motion and boundary state. We represent motion directly in the robot’s configuration space, including full pelvis orientation. A fixed yaw anchor R_A is removed once for a motion sequence, and every frame stores absolute orientation in that anchor:

$$R_t^A = R_A^\top R_t^{\text{world}}, \quad (4)$$

$$x_t = [\Delta p_{xy,t}^{\text{local}}, h_t, \rho(R_t^A), \theta_t]. \quad (5)$$

Here $\Delta p_{xy,t}^{\text{local}}$ denotes planar displacement, h_t is pelvis height, ρ is the six-dimensional rotation representation [38], and θ_t contains 29 joint angles, giving 38 motion dimensions. This retains roll and pitch without multiplying a sequence of predicted rotation increments. Planar displacements are integrated using the appropriate preceding heading. The anchor remains fixed over the generated sequence; it is not reset at every commit.

The corresponding 71-dimensional boundary state is

$$b_t = [h_t, v_{xy,t}^{\text{local}}, v_{z,t}, \rho(R_t^A), \omega_t, \theta_t, \dot{\theta}_t]. \quad (6)$$

The state contains pelvis height, local planar and vertical velocities, orientation, angular velocity ω_t , joint angles, and joint velocities. Conditioning on these quantities makes the starting pose and motion explicit at each segment boundary. They are reconstructed from the committed reference.

Complementary latent roles. A boundary-conditioned encoder maps motion to a 16-dimensional latent per step:

$$z_i = e(q_i) + r_i, \quad q_i \in \{1, \dots, 512\}, \quad r_i \in \mathbb{R}^{16}. \quad (7)$$

Here $e(q_i)$ is the selected codebook embedding. A shared decoder $D_\psi(z, b)$ maps their sum to native motion. The discrete code supplies the structural hypothesis, and the residual adjusts the motion within that

hypothesis. Their complementary roles are encouraged by the reconstruction objectives below; both contribute through the same decoder.

Training the division of labor. Let Φ_f contain native motion, boundary-aware motion velocities, forward-kinematics positions, and their velocities, each standardized using training data. In the native-motion and velocity blocks, the structural view Φ_s selects root quantities, both legs, and the waist; the complementary detail view contains the 14 arm-joint coordinates. In the geometry blocks, structure selects the torso, both wrist endpoints, and both sole points, while detail contains the remaining body points. Positions and their velocities use the same point partition. This couples body organization to endpoint placement: for example, wrist position is structural even though the corresponding arm-joint articulation lies in the complementary native view. The detail view is the exact complement within each block. Equal block weighting avoids letting a high-dimensional geometry block dominate.

Let sg denote stop-gradient. For encoder output z , embedding e , and $r = z - \text{sg}(e)$, we form full and structure-only reconstructions, together with an auxiliary residual-corrupted reconstruction:

$$x_f = D_\psi(\text{sg}(e) + r, b), \quad (8)$$

$$x_s = D_\psi(z + \text{sg}(e - z), b), \quad (9)$$

$$x_c = D_\psi(\text{sg}(e) + \mathcal{C}(r), b). \quad (10)$$

The full branch is supervised on Φ_f ; the other two are supervised on Φ_s . The corruption $\mathcal{C}$ either zeros detail or adds small channel-scaled noise, encouraging the structural reconstruction to persist under changes in the residual. Figure 2 shows the two principal reconstruction paths. The combined objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_f + \mathcal{L}_s + 0.25\mathcal{L}_{\text{robust}} + \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{boundary}}. \quad (11)$$

The final term supervises boundary-transition derivatives and pelvis orientation at the beginning of the reconstructed segment. Appendix A provides the training details.

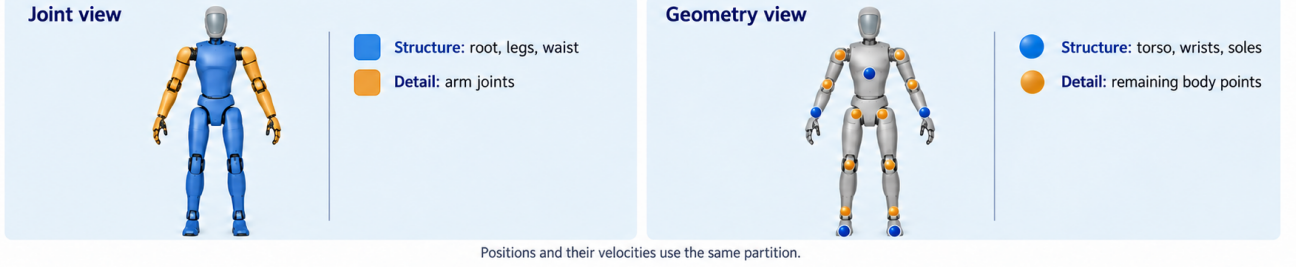
Coupled generation. A causal Transformer samples structural tokens autoregressively. A continuous diffusion generator [9] conditions on the sampled structure, history, state, and forecast music to predict residual detail. We use ten DDIM steps [28]. The structural planner uses a classification objective, and the detail generator uses a denoising objective. Conditioning detail on the sampled structure connects their decisions before decoding. Committing the same prefix of both branches preserves that pairing across successive plans.

4.3 Commit Forcing

A committed motion segment changes two inputs to the next decision: the latent history and the boundary state. Under TF, both are taken from recorded motion; under streaming inference, both depend on the model’s

Kinematics-guided structure-detail decoupling

(a) Body-part supervision views



(b) Two principal reconstruction paths

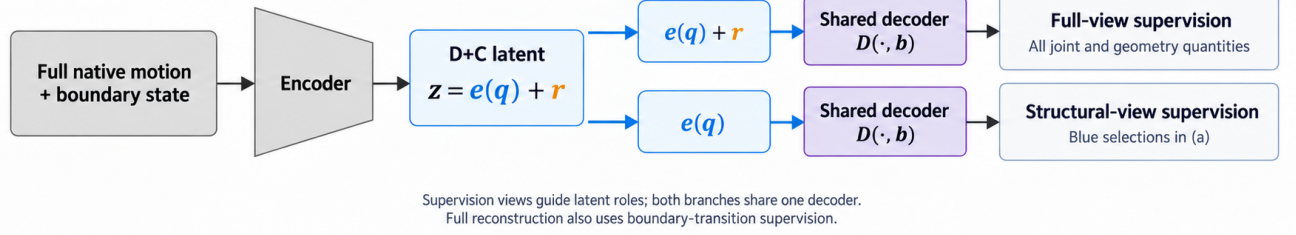


Figure 2: The representation and its training are one design. Body-part supervision has separate joint and geometry partitions: root/legs/waist versus arms in joint space, and torso/wrists/soles versus other body points in geometry. The two principal reconstruction paths supervise the full latent on all views and discrete structure on the structural view. The shared decoder and boundary state are used throughout.

samples. CoF addresses this mismatch by constructing the generated transition before supervising continuation (Fig. 3). A stop-gradient first pass samples $\tilde{q}, \tilde{r}$ using the inference samplers. Their first C steps are committed together and decoded; an update operator U appends the codes and derives the new state:

$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \quad (12)$$

The next supervised plan receives this complete context. Substituting the whole transition avoids pairing generated codes with a boundary state taken from a different recorded transition.

Residual targets under sampled structure. For the next dataset target $z^* = e(q^*) + r^*$ and sampled structural token $\hat{q}$, define

$$r_{\text{rebased}}^* = z^* - e(\hat{q}). \quad (13)$$

Thus $e(\hat{q}) + r_{\text{rebased}}^* = z^*$. This preserves the *target latent*; the boundary-conditioned decoder need not produce identical motion under recorded and generated states. We retain the next recorded latent target while changing the conditioning history and state. Each training example constructs its first-pass transition independently; Appendix A describes this sampling procedure.

The key operation is the joint transition: generated structure, generated detail, and reconstructed boundary state enter the next training context together. This exposes the continuation model to the consequences of its own commitment operation. Boundary continuity and long-rollout quality provide the corresponding tests against TF.

Supporting training choices. Full-history corruption (FHC) perturbs the complete discrete-embedding and residual history during supervised training, while leaving boundary state and forecast music unchanged. It is applied after CoF context construction and is absent at inference. This regularizes the history input across the complete context window. We consider two startup settings—mixed song-origin starts and ordinary continuation—and report their effects separately in the experiments.

4.4 Robot Reference Integration and Execution

The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [23]. A reference bridge reconstructs the root trajectory from the native representation in Eq. (5), maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. For the current representation, planar displacement is integrated using the preceding heading, while pelvis orientation is recovered from the absolute rotation within the fixed sequence yaw anchor. The bridge does not accumulate a sequence of predicted yaw increments. Only committed motion is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions. The generator separately updates its latent history and boundary state from committed reference motion. These are distinct feedback paths: generated references condition subsequent planning, while measured execution closes the controller loop. The inte-

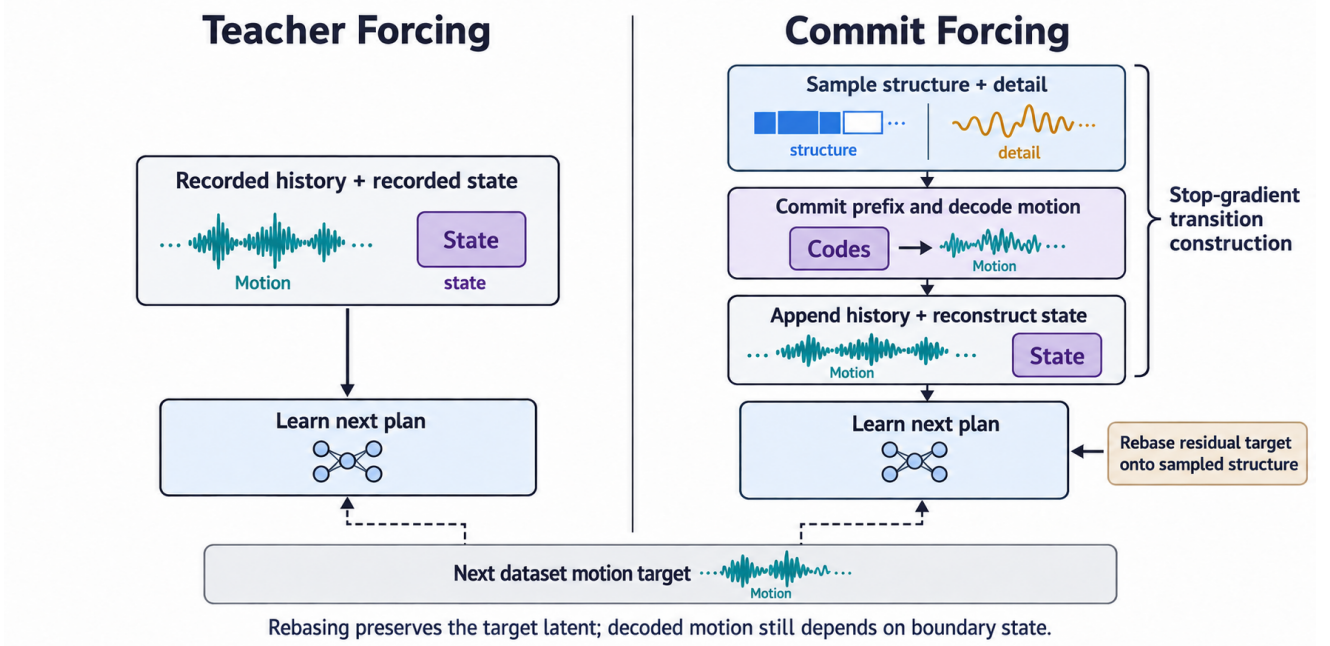


Figure 3: Learning the next plan: TF versus CoF. TF receives a recorded history–state pair. CoF samples and commits structure and detail together, derives the corresponding next reference state, and supervises continuation. Both use recorded targets. Residual rebasing retains the target latent under sampled structure; decoding additionally depends on the supplied boundary state.

gration therefore connects the reference stream to motion tracking without changing the generator into a measured-state feedback policy. Appendix D specifies the delivery and measurement protocol and separates earlier interface diagnostics from evaluation of the current generator.

5 Experiments

Our evaluation centers on three questions: whether forecast conditioning adds value beyond heard music, whether kinematic supervision produces complementary structure and detail, and whether learning from committed transitions improves continuation over TF. We report completed music-condition and history-training comparisons on FineDance and specify the matched component studies whose results are pending.

5.1 Data, Controls, and Metrics

We use FineDance motion retargeted to G1 and represented as in Eq. (5). The quantitative cohort comprises 18 official test tracks, with three independently trained models and three sampling seeds per model. Each track contributes the same source-supported 23 s segment across compared methods. The 60 s qualitative sequence is evaluated separately from this quantitative cohort. Startup training is either *mixed-start* (song-origin starts mixed with ordinary windows) or *continuation-start* (ordinary continuation windows).

Music matching. Our frozen G1 music–motion retrieval evaluator, inspired by SoulNet’s matching formulation [18], scores compatibility between audio and

generated motion. Its MMR-MS distance is lower for better matching; it combines sequence-level and temporal features. The evaluator is independent of generator optimization. We report this as the primary matching measure, accompanied by beat alignment (BAS). These complementary measures assess sequence compatibility and rhythmic correspondence, respectively.

Motion quality and continuation. Kinetic and geometric Fréchet distances measure distribution similarity; their corresponding diversity measures describe motion variation. Root motion, contact/grounding proxies, and abrupt changes characterize properties that distribution scores do not resolve. The CoF comparison is designed to assess boundary continuity and all three non-overlapping 20 s portions of a 60 s rollout. Appendix B details the paired evaluation and uncertainty estimates.

5.2 Anticipatory Music Conditioning

Does the generator use its music forecast? A forecast is useful to the dance generator only if its information changes the resulting music–motion correspondence. Table 1 compares forecast-conditioned training with separately trained no-music controls. Mixed-start training reduces MMR-MS by 0.782, with a 95% interval of [0.294, 1.285]; continuation-start training reduces it by 1.123, with an interval of [0.621, 1.649]. All three training seeds agree in direction under both settings. The agreement across startup settings supports the use of forecast-derived music information beyond a particular initialization policy.

Table 1: Music-condition use. Reduction in MMR-MS relative to matched no-music training; positive is favorable. Paired hierarchical 95% intervals use the complete 18-track cohort.

Startup	Improvement	95% interval
Mixed-start	0.782	[0.294, 1.285]
Continuation-start	1.123	[0.621, 1.649]

Isolating anticipation. The stronger question is whether the forecast contributes beyond information already present in heard music. The matched design in Table 2 changes the music states while holding representation, CoF, conditioning capacity, training budget, history policy, and sampling settings fixed. Wrong-song conditioning provides a complementary specificity control. Results for the matched heard-versus-predicted comparison are pending; the completed no-music comparison establishes condition use.

5.3 D+C Representation and Training

Complementary roles. The structural objective is intended to preserve body organization while allowing complementary articulation. To test this mechanism, the study design combines reconstruction in the two kinematic views with swaps of discrete and continuous codes. A stronger structural response to a discrete-code swap than to a detail-code swap would support the intended division of labor. Reconstruction quality, residual contribution, and effective rank must be considered together so that an apparently selective response is not explained by a collapsed branch. These mechanism results remain pending.

Generation benefit. Table 3 specifies the complementary generation test: separately trained discrete-only and continuous-only generators against D+C with its paired supervision. The comparison uses the same full-orientation motion representation and matched training and evaluation settings. Unlike zeroing a residual at inference, separately trained alternatives test how each representation supports generation when optimized in its own right. Results are pending.

5.4 Commit Forcing versus Teacher Forcing

The TF comparison isolates the consequence of learning from a complete generated transition. Its design holds music conditions, D+C codec, startup, history corruption, optimization budget, and inference sampler fixed, changing how the next training history and state are constructed. The intended evaluation spans 60s, reporting distribution and diversity alongside boundary statistics and all three 20s positions. This tests whether a benefit persists as generated context accumulates instead of being confined to the start of a rollout. Matched TF/CoF results are pending (Table 4).

5.5 Effect of History Corruption

We compare clean-history training with FHC to examine how regularizing the motion context affects the generated distribution. Under mixed-start training, FHC reduces kinetic/geometric FID from 208.599/0.482 to 162.831/0.410, with favorable directions across all training seeds. The continuation-start changes are smaller, from 156.218/0.372 to 154.180/0.361. This shows that the distribution-level effect depends on the startup policy.

The matching and physical proxies give a more specific interpretation. MMR-MS point estimates improve under both startup settings, but the paired intervals for FHC versus clean history include zero. Mixed-start contact and jerk measures favor clean history. FHC therefore changes the distribution-smoothness tradeoff; its distribution gains do not establish a separate gain in musical matching. Appendix C reports the complete metric values.

5.6 Long-Sequence Generation

Figure 4 illustrates one 60s ForeDance generation. Snapshots at 10, 30, and 50s show the evolving reference poses across the stream. The sequence demonstrates the output format and provides a qualitative view of one generated dance; it is not a comparative estimate of long-horizon quality.

The native references include the configuration and motion information needed by a separate whole-body tracker. The present illustration evaluates reference generation; a matched current-model controller-execution case remains pending. Section 5.7 separately examines the execution interface.

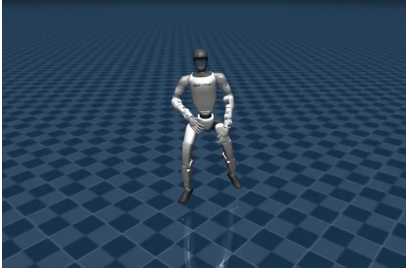
5.7 Reference Execution with a Fixed Controller

The execution interface asks whether the delivered reference retains its motion characteristics after tracking. We distinguish the generated motion, the bridge’s sent reference, the controller’s consumed target, and measured execution. This separates changes introduced by representation conversion and delivery from target-tracking error.

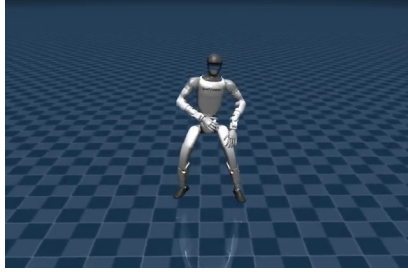
A separate MuJoCo replay study of an earlier 34-dimensional generator configuration provides supporting measurements of this interface (Appendix D). Four previously generated trajectories were replayed through the same fixed SONIC controller with the same reference-delivery settings. The reported motion-amplitude and velocity-variation measurements show why tracking should be examined beyond joint-position error. Because these are fixed-reference replays, their comparison addresses tracking of the supplied motion, not online music inference during execution. They are also separate from the current 38-dimensional ForeDance cohort. A matched recording of the current generated reference and its execution remains to be evaluated; the rendered sequence in Fig. 4 is reference motion.

Table 2: Forecast versus heard-music conditioning. Matched comparison design; results are pending (—). Each metric uses identical sources and sampling conditions across methods.

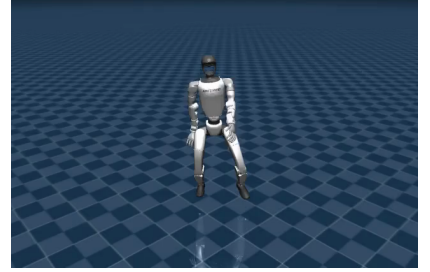
Condition	MMR-MS ↓	BAS ↑	FID _k ↓	FID _g ↓	Div _k	Div _g
Heard-music states	—	—	—	—	—	—
Predicted-music states	—	—	—	—	—	—



10 seconds



30 seconds



50 seconds

ForeDance: selected 60-second FineDance generation.

Figure 4: A 60-second ForeDance generation. Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

Table 3: Representation and training comparison. Matched generator results are pending (—).

Representation	FID _k ↓	FID _g ↓	Div _k	Div _g
Discrete only	—	—	—	—
Continuous only	—	—	—	—
D+C with paired training	—	—	—	—

Table 4: Commit Forcing versus Teacher Forcing. Matched 60 s rollout results are pending (—).

Training	FID _k ↓	FID _g ↓	Div _k	Div _g
Teacher Forcing	—	—	—	—
Commit Forcing	—	—	—	—

6 Discussion

Anticipation as a conditioning interface.

ForeDance makes predicted musical context available before the corresponding audio is observed. This is useful at the interface between musical planning and irreversible motion commitment: each forecast guides a short plan while subsequent decisions incorporate newly arrived music. The no-music comparisons show that the generator uses this forecast-derived information under both startup settings. Their scope is music-condition use; determining the incremental benefit over heard-music features requires the matched forecast comparison.

Connecting motion organization and continuation.

The structural objective expresses body organization in both joint and Cartesian views, allowing arm articulation to vary while constraining wrist placement. Shared decoding then combines these views into one motion trajectory. CoF carries that coupling across decisions by deriving the next boundary state from the same structure–detail prefix that enters the history. The representation and continuation comparisons re-

main pending, so these mechanisms motivate the design without yet establishing their separate quantitative gains.

Scope of the evaluation. The current-generator numerical results use the complete fixed cohort of 18 FineDance test tracks; the long-sequence illustration is a selected case. The reported intervals reflect training-seed and track variation under the paired protocol. Distribution, diversity, matching, and contact measure different properties: the FHC results illustrate a distribution–smoothness tradeoff rather than a uniform improvement. The primary study evaluates robot-native reference generation. The separate four-run MuJoCo diagnostics in Appendix D concern an earlier generator configuration and fixed-reference tracking; they are not pooled with the current generation cohort. The rendered current-model example does not establish controller execution, and the generation schedule alone does not establish computational throughput.

From reference generation to execution. The bridge connects committed motion to a fixed whole-body controller while keeping reference-derived planning state distinct from measured execution state. The supporting replay diagnostics show that reference amplitude and motion dynamics can be retained to different degrees, so joint-position error alone does not describe the executed motion. Their scope is the supplied references in simulation. Assessing the current generator through this interface requires a paired current-model reference and execution recording.

7 Conclusion

We introduced ForeDance, a framework that connects causal music forecasting with structured streaming dance generation. Predicted music states guide

discrete structure and continuous detail; kinematic reconstruction objectives define their complementary roles; and Commit Forcing trains continuation through jointly committed motion and its reconstructed state. FineDance results demonstrate improved music–motion matching over matched no-music training under two startup settings. Together, the method and these findings support a concrete interface for using musical forecasts while generating an evolving dance from committed motion. A separate reference bridge connects this output to fixed whole-body tracking, with execution diagnostics kept distinct from the current generator evaluation.

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A Representation and Training Details

Motion representation. We use 38-dimensional native motion and a 71-dimensional boundary state. Pelvis orientation is absolute within one fixed yaw anchor for the sequence. The encoder and decoder are boundary-conditioned, and the next state is reconstructed from committed reference motion. The codebook has 512 embeddings and a 16-dimensional continuous residual. At 30 Hz and two frames per latent step, the generator retains 64 history steps, plans eight steps, and commits four steps.

Kinematic reconstruction losses. The full view contains four blocks: native motion, native velocity, forward-kinematics positions, and geometry velocity. Each block is standardized with training-split statistics. For a view Φ with blocks $\Phi^{(k)}$, define the reconstruction error

$$\mathcal{E}_{\Phi}(\hat{x}, x) = \frac{1}{4} \sum_{k=1}^4 \text{mean} \left[\left(\Phi^{(k)}(\hat{x}) - \Phi^{(k)}(x) \right)^2 \right]. \quad (14)$$

This gives $\mathcal{L}_f = \mathcal{E}_{\Phi_f}(x_f, x)$, $\mathcal{L}_s = \mathcal{E}_{\Phi_s}(x_s, x)$, and $\mathcal{L}_{\text{robust}} = \mathcal{E}_{\Phi_s}(x_c, x)$. Equal block weights prevent the number of geometric coordinates from determining their relative contribution. The VQ term is the codebook loss plus 0.25 times the commitment loss.

For the auxiliary robustness objective, the residual is zeroed with probability 0.5. Otherwise, Gaussian noise is added with scale sampled uniformly from 0.01–0.05 times the residual channel standard deviation. The resulting reconstruction is supervised on the structural view. This codec objective is separate from FHC, which regularizes the generator’s motion history.

The boundary loss supervises the first three decoded frames relative to the supplied boundary. It combines normalized smooth- L_1 errors for velocity, acceleration, and jerk with weights 1, 0.5, and 0.25, respectively, plus pelvis-rotation geodesic error with weight 1. These quantities explicitly penalize a reconstructed segment whose beginning is inconsistent with the target transition.

Body-part supervision views. In the native block, structure selects the first 24 of 38 coordinates: nine root coordinates, twelve leg joints, and three waist joints. The remaining 14 coordinates are the bilateral shoulder (three per arm), elbow (one), and wrist (three) joints. In the 35-dimensional velocity block, structure selects the first 21 coordinates: root linear/angular velocity and the same 15 leg/waist joint velocities. The other 14 are arm-joint velocities.

The geometry block contains 31 points: 29 non-root body origins and two lowest-foot geometry points. Structure selects the torso, the two wrist-yaw body origins, and the two lowest-foot points. Detail selects the remaining 26 points. Geometry velocity uses the same partition. These are supervision and analysis views, not independent anatomical decoders. In particular, the structural geometry view can constrain wrist placement while the complementary native view describes arm articulation.

Generator optimization and initialization. Each reported training run uses 300,000 windows, batch size 48, and 6,250 updates. We train three seeds and sample each model with three seeds. Forecast states enter both branches through frame-wise modulation; the music model remains frozen. The residual sampler uses ten DDIM steps with guidance fixed at 1.0. Mixed-start training draws 20% song-origin/prefix starts and 80% ordinary windows. Continuation-start training uses ordinary windows throughout.

History regularization. FHC applies to the complete 64-step history with probability 0.5 per example. The discrete embeddings and continuous residual history share a severity timestep sampled from 850–999, with independent Gaussian noise. Their masks and timestamps are preserved, as are the boundary state and forecast conditions. Corruption is applied in the supervised pass after CoF context construction and is disabled at inference.

Constructing CoF transitions. Each training example independently samples a first-pass transition from its supplied context. The committed discrete and residual prefixes enter the history together, and their decoded motion determines the next boundary state. The next recorded structural and latent targets remain fixed. Residual rebasing expresses that latent target relative to the sampled structure; it does not re-encode the target under the generated boundary. This construction trains local continuation after a generated commit without requiring a fully recursive long training rollout.

B Evaluation Protocol

The current-generator numerical tables use all 18 tracks in the fixed FineDance test cohort. The common source-supported segment is 23 s, beginning at source frame 264. Each method has three training seeds and three sampling

seeds per training run, giving 162 paired track-level outputs. Methods within a comparison share source identities, segment boundaries, and sampling settings.

For the music-condition comparison, the no-music controls are separately trained models. They are not obtained by removing the condition from a forecast-trained model at inference. This controls for the learning procedure associated with the absence of music, while preserving the distinction from the pending heard-music comparison.

MMR-MS is a frozen learned matching distance using independently trained G1 motion features. It follows the alignment motivation of SoulNet but uses a robot-motion evaluator rather than the original human-motion checkpoint. Feature extraction and normalization are held fixed throughout the comparisons. Lower MMR-MS denotes better matching; beat alignment supplies a complementary rhythmic measure.

We compute paired 95% intervals using 10,000 hierarchical bootstrap draws. Each draw resamples training seeds with replacement, then tracks within each sampled training seed. The repeated sampling-seed outputs are averaged within each track. This retains training variation while avoiding the treatment of individual motion frames as independent experimental samples.

The reported current-generator numerical tables use the complete cohort without favorable-track filtering. Figure 4 is a selected qualitative example from a separate 60s rollout. Three original frames are taken at 10, 30, and 50s from the same generated trajectory. The comparison layouts in the main text identify studies with pending results; they do not contribute to the reported estimates or intervals.

C History-Corruption Results

Table 5 compares FHC with clean-history training under both startup settings. In mixed-start training, FHC reduces both distribution distances and increases kinetic diversity, while PFC and jerk favor clean history. Under continuation-start training, distribution changes are smaller and the diversity measures move in different directions. The FHC-versus-clean-history intervals for MMR-MS include zero in both settings. The distribution changes therefore inform the choice of history regularization without establishing a corresponding improvement in music matching.

Table 5: Effect of history corruption on the full 18-track cohort. Lower is favorable for MMR-MS, FID, and PFC. Diversity describes motion spread and is interpreted together with fidelity.

Startup	History	MMR-MS	FID _k	FID _g	Div _k	Div _g	PFC
Mixed-start	Clean history	40.902	208.599	0.482	7.495	0.493	0.143
Mixed-start	History corruption	40.859	162.831	0.410	8.626	0.508	0.167
Continuation-start	Clean history	41.195	156.218	0.372	8.329	0.545	0.163
Continuation-start	History corruption	41.029	154.180	0.361	9.017	0.541	0.162

D Reference Delivery and Supporting Execution Diagnostics

Study identity and scope. The execution measurements below come from a separate study of the earlier 34-dimensional native-motion generator and its SONIC reference bridge. They are retained as supporting interface diagnostics and are not results for the current 38-dimensional ForeDance model. This study uses one FineDance source track and one sampling seed, with four generated reference trajectories replayed in MuJoCo. It is separate from the 18-track, three-training-seed generation evaluation in Appendix B; its runs are not pooled with that cohort or its uncertainty estimates. Each replay lasts approximately 60s. Music inference does not run during replay, including for the trajectory originally generated with live forecast conditions.

Reference construction and delivery. A four-step latent commit contains eight 30 Hz frames. A continuous 50 Hz time grid yields 13 or 14 reference samples per commit without resetting the sampling phase at each packet. The controller receives committed references rather than motor torques. The four diagnostic replays use the same fixed SONIC controller at real-time playback, with no uncommitted-plan preview, clipping, startup ramp, or alignment hold. Initial state, delivered future-reference access, and packet timestamps are part of the protocol; full-trajectory delivery and committed-prefix delivery are not interchangeable controls. The trajectories begin at an artificial continuation point with empty motion history, rather than at full-song onset.

Observed layers. Let M_{gen} denote decoded generator motion, M_{sent} the resampled reference sent by the bridge, M_{target} the controller’s consumed reference, and M_{exec} measured execution. Joint ordering, coordinate conventions, default offsets, and timestamps must agree before comparing these layers. The archived diagnostic summaries use named target and measured fields in MuJoCo joint order. This accounting distinguishes conversion, delivery, and tracking discrepancies. It does not imply that measured state is fed back into the generator.

Conditions and measurements. All four conditions use inferred musical forecasts, never ground-truth future audio. The first three share cached forecast content but differ in its consumption-age trace: an ideal-age reference, the earlier measured trace, and a revised measured trace. The fourth uses live forecast content with the revised trace. These labels identify how a trajectory was generated before replay. They do not turn a tracking comparison into a measure of musical quality.

For active joint j , amplitude and velocity-variation retention are

$$R_j^{\text{amp}} = \frac{P_{95}(q_j^{\text{exec}}) - P_5(q_j^{\text{exec}})}{P_{95}(q_j^{\text{target}}) - P_5(q_j^{\text{target}})}, \quad (15)$$

$$R_j^{\text{dyn}} = \frac{\mathbb{E}[(\dot{q}_j^{\text{exec}})^2]}{\mathbb{E}[(\dot{q}_j^{\text{target}})^2]}. \quad (16)$$

Only joints with a target percentile range above 0.02 rad enter the active-joint summary, and ratios with near-zero denominators are excluded. The reported retention values are medians of joint-wise ratios. The dynamic ratio measures squared joint velocity; it is not physical energy or a whole-body energy ratio. Joint-position RMSE is evaluated after time alignment. Alignment lag is estimated within ± 0.5 s on the 50 Hz grid; the four recorded estimates are 40 ms, which is a case-specific alignment result rather than a general controller or system latency. Steady-state summaries exclude the first two seconds, so they do not establish startup behavior.

Table 6: Supporting fixed-reference replay measurements for an earlier generator configuration. Each row is one MuJoCo replay, excluding its first two seconds. Rows differ in the previously generated reference, not in the fixed SONIC controller. These measurements do not compare current ForeDance variants.

Reference-generation condition	Aligned RMSE (rad)	Median R^{amp}	Median R^{dyn}
Cached forecast, ideal age	0.1421	0.869	0.496
Cached forecast, earlier age trace	0.1281	0.858	0.428
Cached forecast, revised age trace	0.1394	0.877	0.412
Live forecast, revised age trace	0.1431	0.901	0.411

Interpretation. Across these four references, the reported amplitude retention lies between 0.858 and 0.901, while squared-velocity retention lies between 0.411 and 0.496. Position tracking therefore coexists with attenuation of motion dynamics in this configuration. This does not identify which attenuated components carry musical expression: high-frequency variation may include both meaningful articulation and jitter. Nor do four single runs establish repeatability, a ranking of music conditions, or real-hardware performance. The measurements motivate paired inspection of target and executed motion; a current-model execution evaluation must retain its own reference identity, initialization, recording coverage, and coordinate checks.